



# Kitchen Object Segmentation & Annotation Project

**Project Name:** Kitchen Object Segmentation Dataset Annotation

**Client:** Private Client (Fiverr)

**Annotation Team:** Shayan Khan, Zeeshan khan, Muhammad Shaheer

**Tool used:** CVAT

## 1. Project Overview

Annotated kitchen images for a computer vision dataset aimed at detecting and segmenting various kitchen components. The project involved creating precise polygon annotations to generate high-quality training data for an AI model.

## 2. Dataset Description

- **Total images:** 500 Images
- **Total classes:** 16
- **Resolutions:**  $1024 \times 1024$
- **source:** given by client
- **Annotation Type:** Polygon Segmentation
  - Classes Annotated
    1. Google Drive
    2. base\_drawers
    3. base\_units
    4. cooker

5. dishwasher
6. extractor
7. fridge
8. hob
9. kitchens
10. microwave
11. oven
12. sink
13. tall\_units
14. tap
15. wall\_units
16. worktop

### **3. Responsibilities**

- Reviewed and organized the image dataset before annotation.
- Created accurate polygon annotations for all kitchen objects.
- Ensured precise object boundaries for segmentation tasks.
- Maintained consistency across all class labels.
- Performed quality checks and corrected annotation errors.
- Delivered completed annotations according to client requirements.
- Challenges
  - Annotating overlapping kitchen appliances and furniture.
  - Maintaining consistent labeling across multiple object categories.
  - Creating detailed polygon masks for complex object shapes.

### **4. Outcome**

Successfully delivered a fully annotated segmentation dataset consisting of 500 images with 15 kitchen-related classes. The dataset was prepared for use

in computer vision model training and object segmentation tasks.

## 5. **Skills Demonstrated**

Polygon Segmentation

Image Annotation

Data Labeling

Quality Assurance

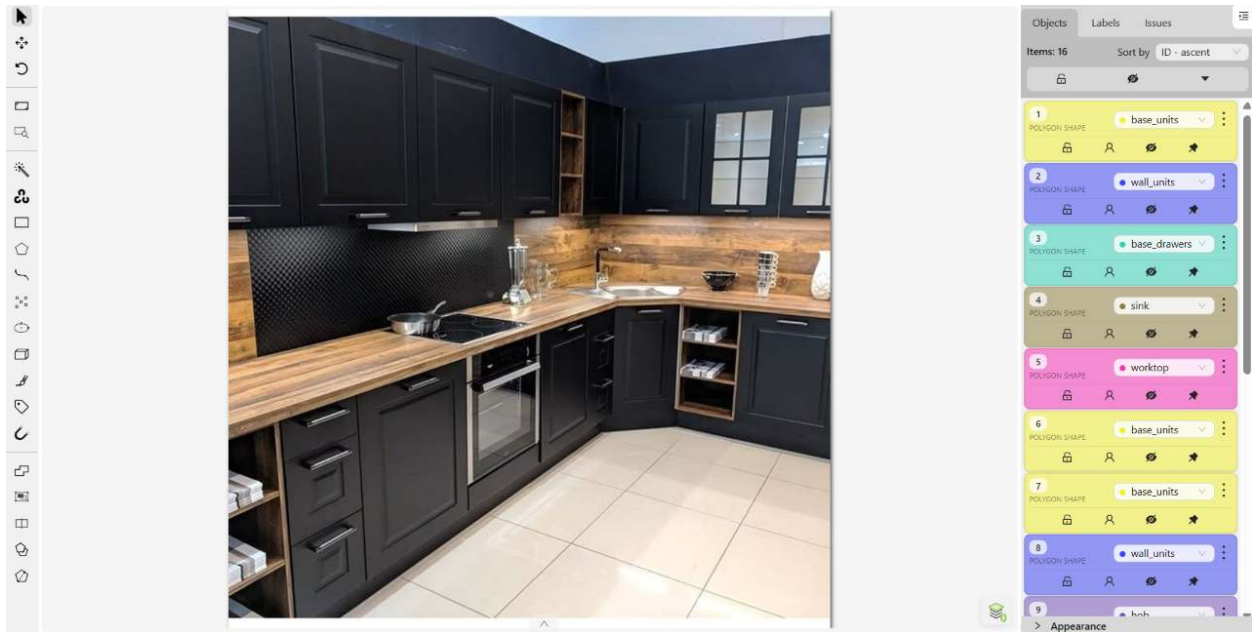
CVAT Annotation

Computer Vision Data Preparation

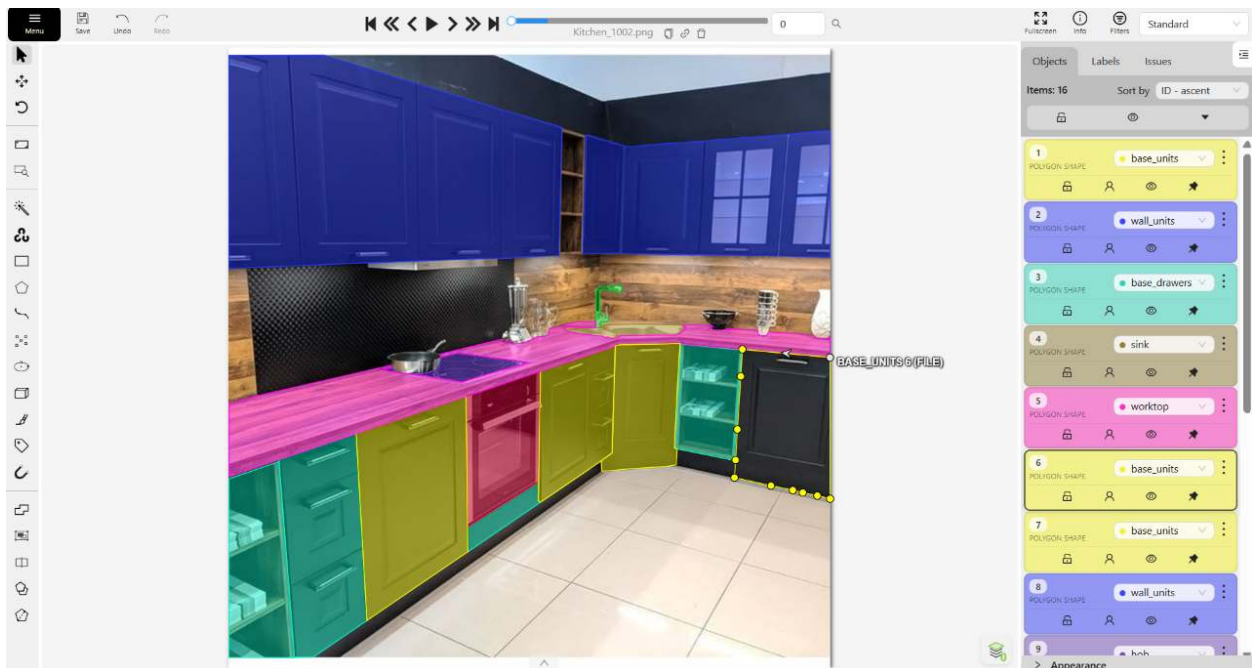
## 6. **Final output:**

Completed **500 kitchen image annotations** with **15 object classes** using polygon segmentation while maintaining annotation quality and consistency throughout the dataset.

### **Example screenshots**



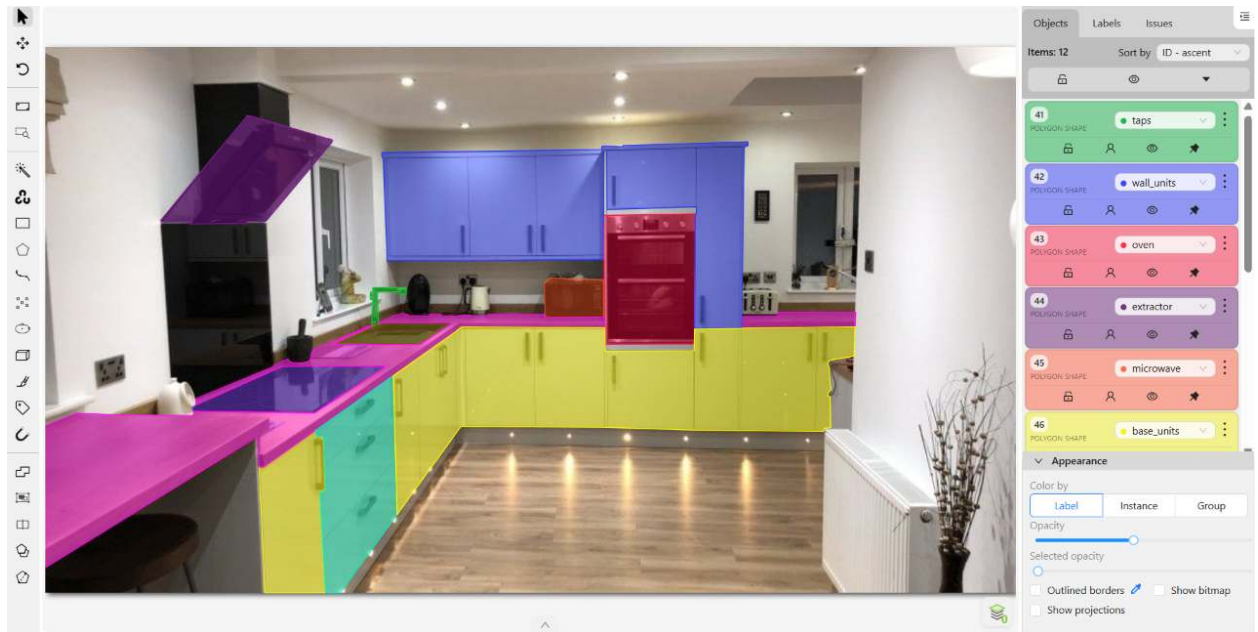
[Fig 5.1]: Before Annotations



[Fig 5.2]: After Annotations



[Fig 5.3]: Before Annotations



[Fig 5.4]: After Annotations

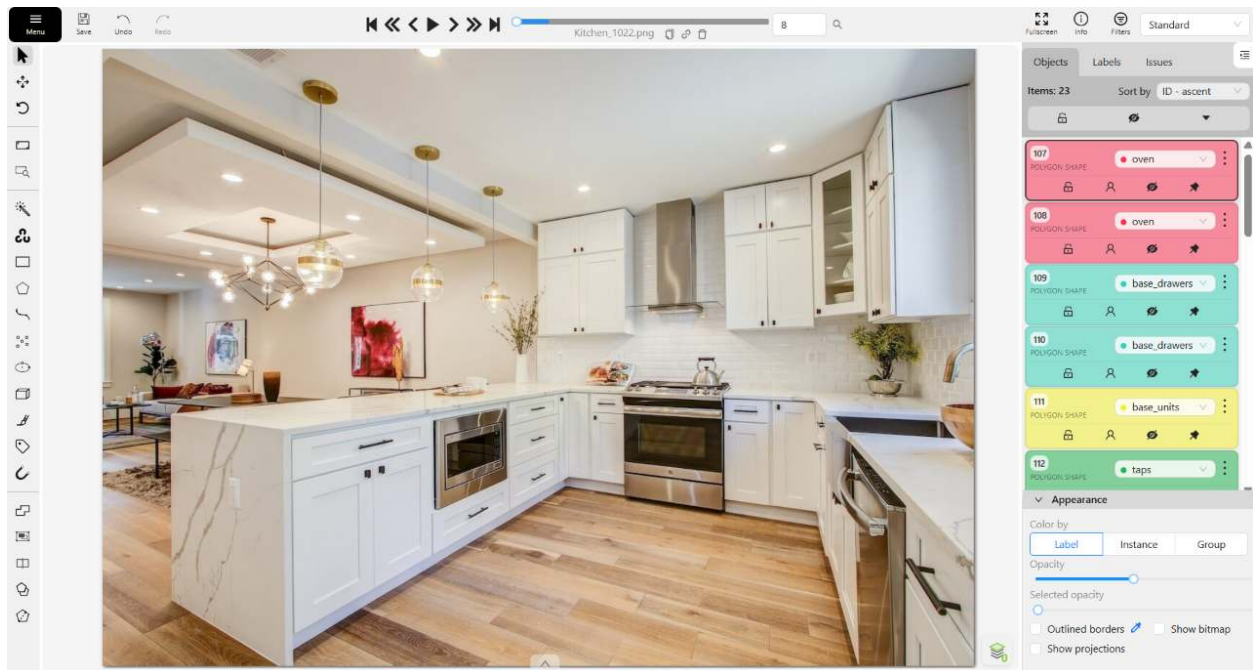


[Fig 5.5]: Before Annotations



[Fig 5.6]: After Annotations

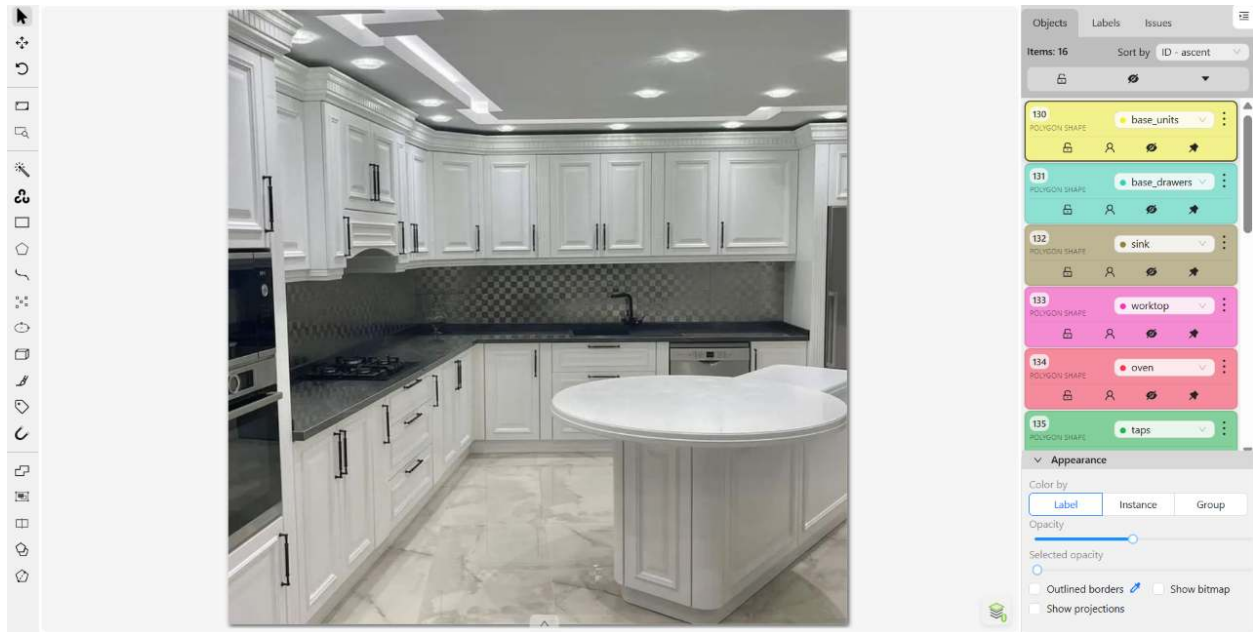




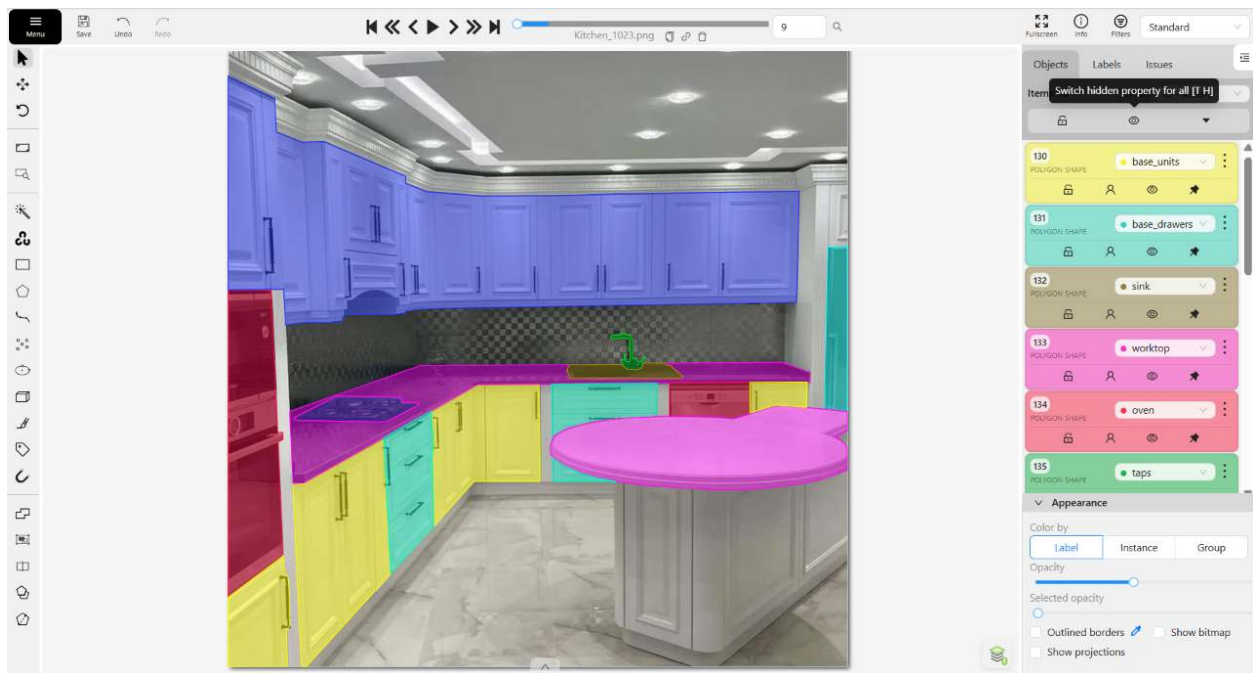
[Fig 5.7]: Before Annotations



[Fig 5.8]: After Annotations

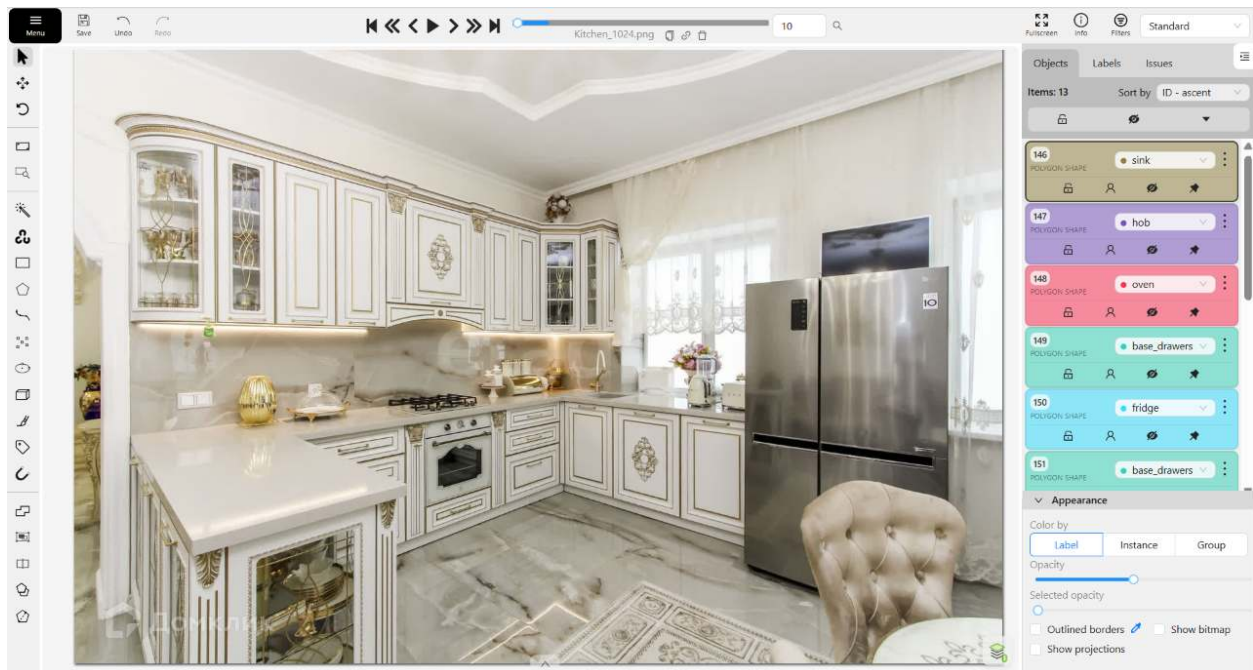


[Fig 5.9]: Before Annotations

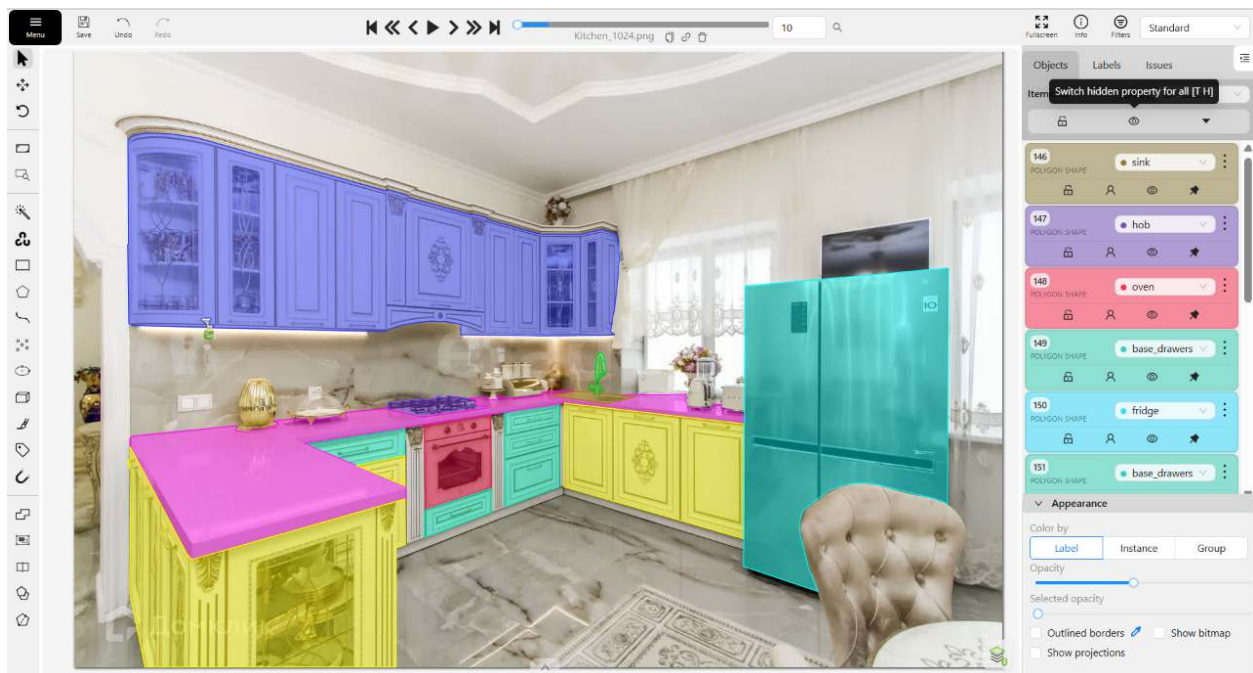


[Fig 5.10]: After Annotations





[Fig 5.11]: Before Annotations



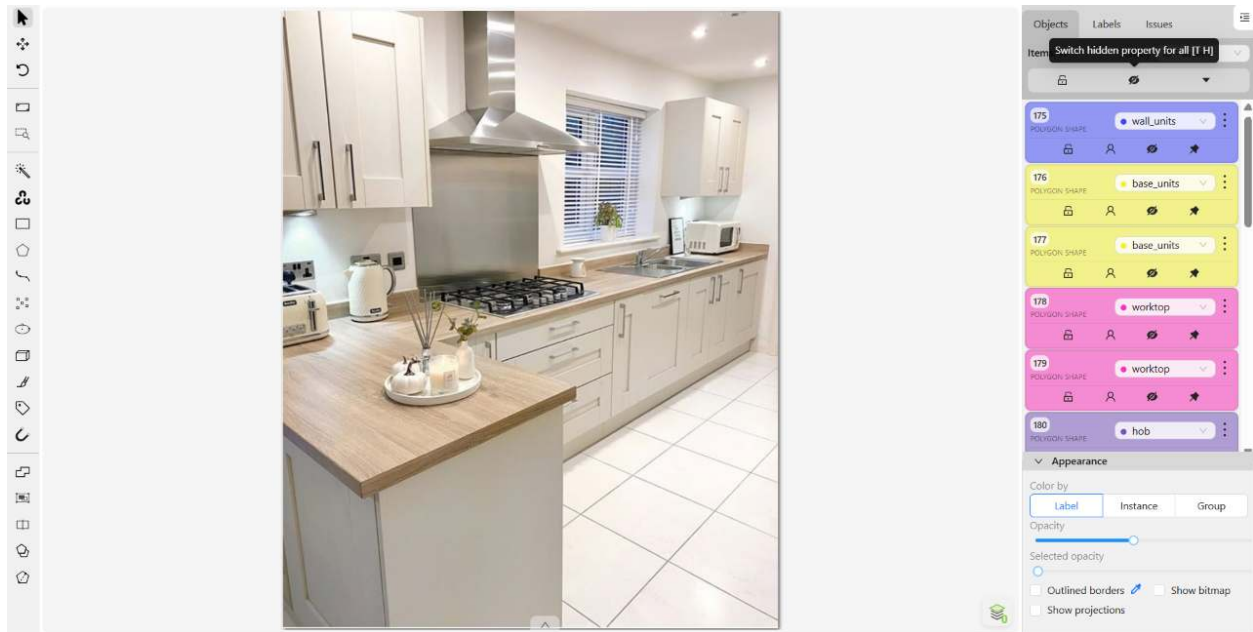
[Fig 5.12]: After Annotations



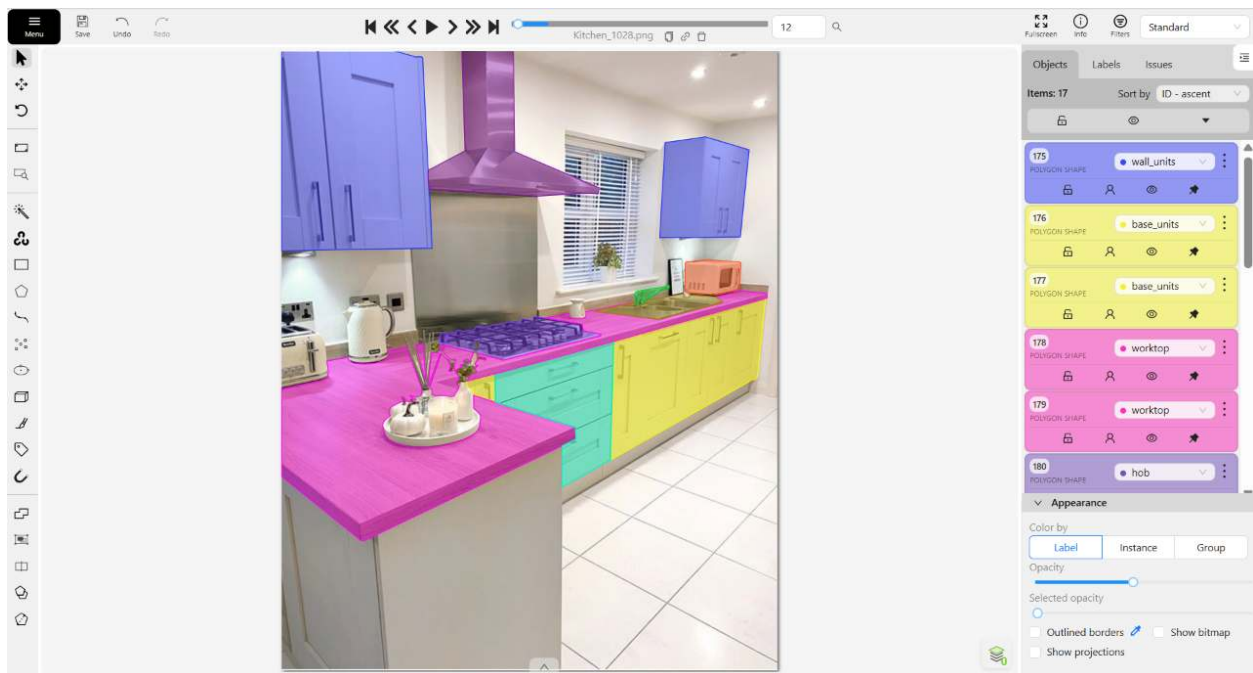
[Fig 5.13]: Before Annotations



[Fig 5.14]: After Annotations

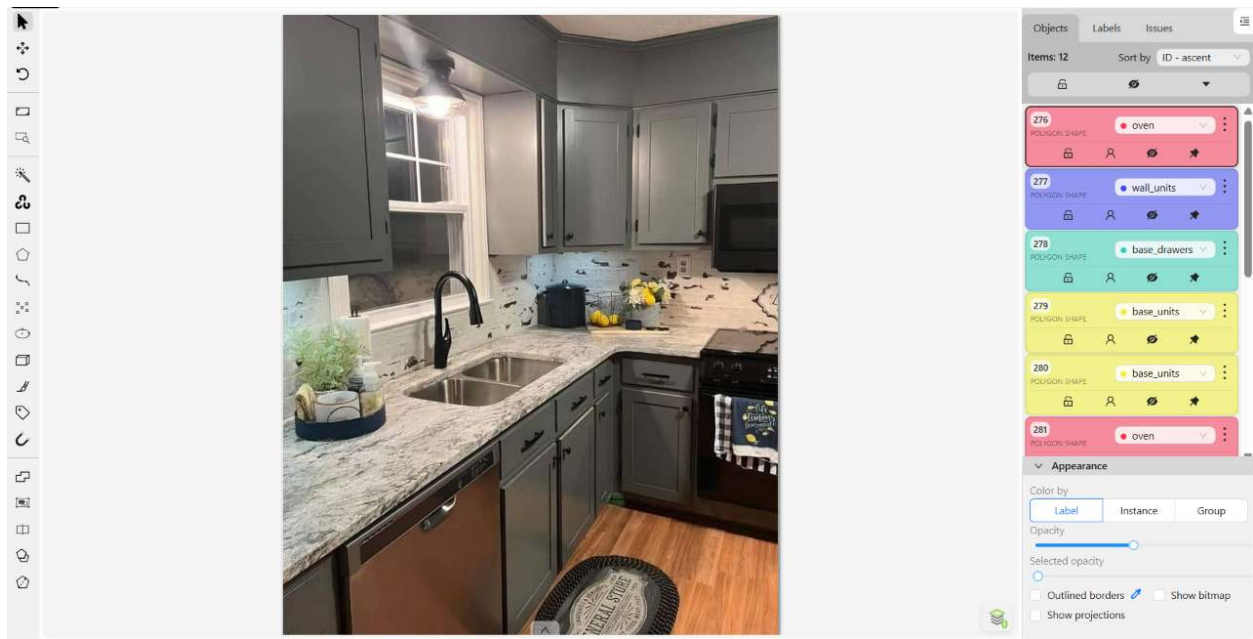


[Fig 5.15]: Before Annotations

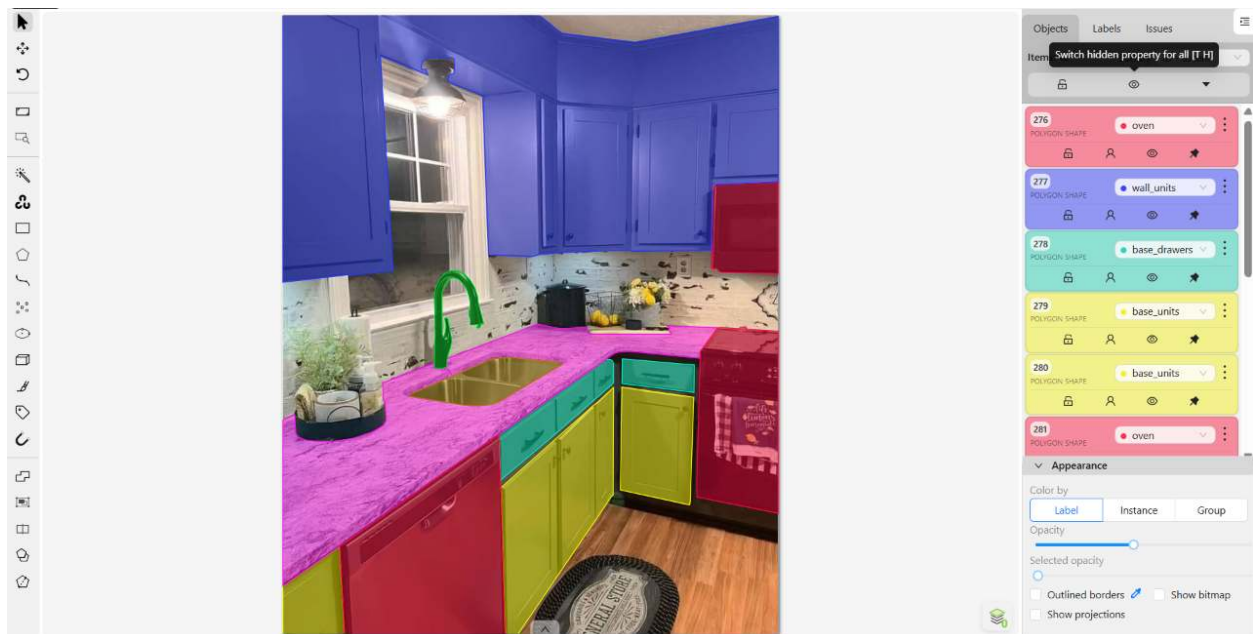


[Fig 5.16]: After Annotations





[Fig 5.17]: Before Annotations



[Fig 5.18]: After Annotations



[Fig 5.19]: BeforeAnnotations

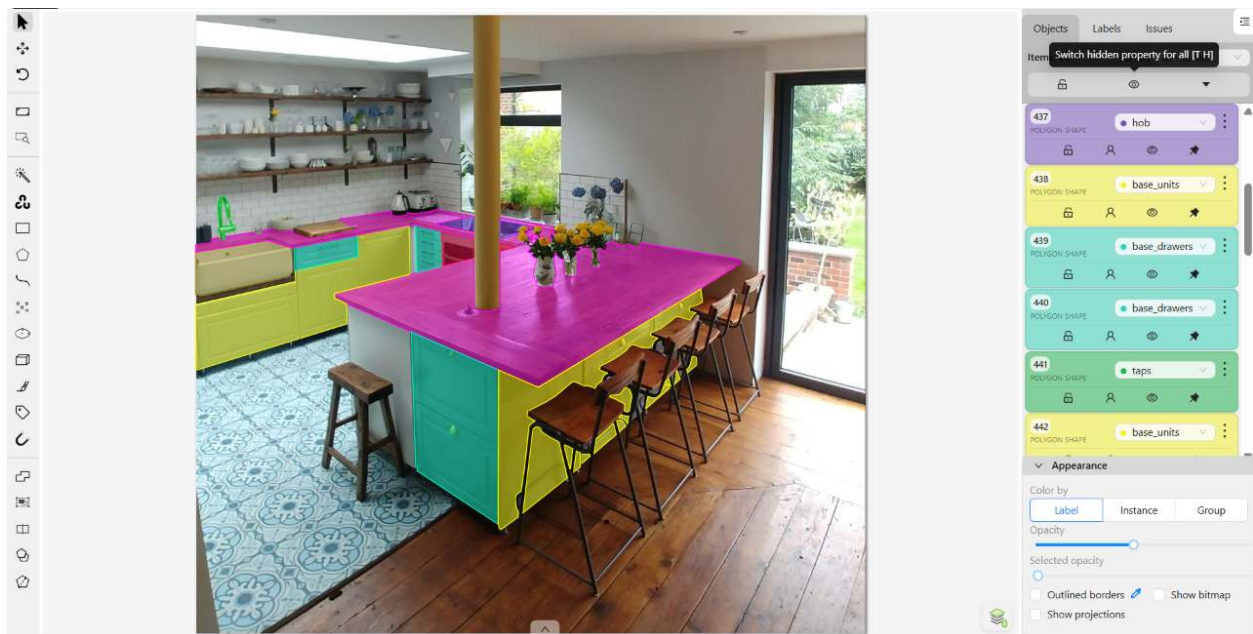


[Fig 5.20]: After Annotations





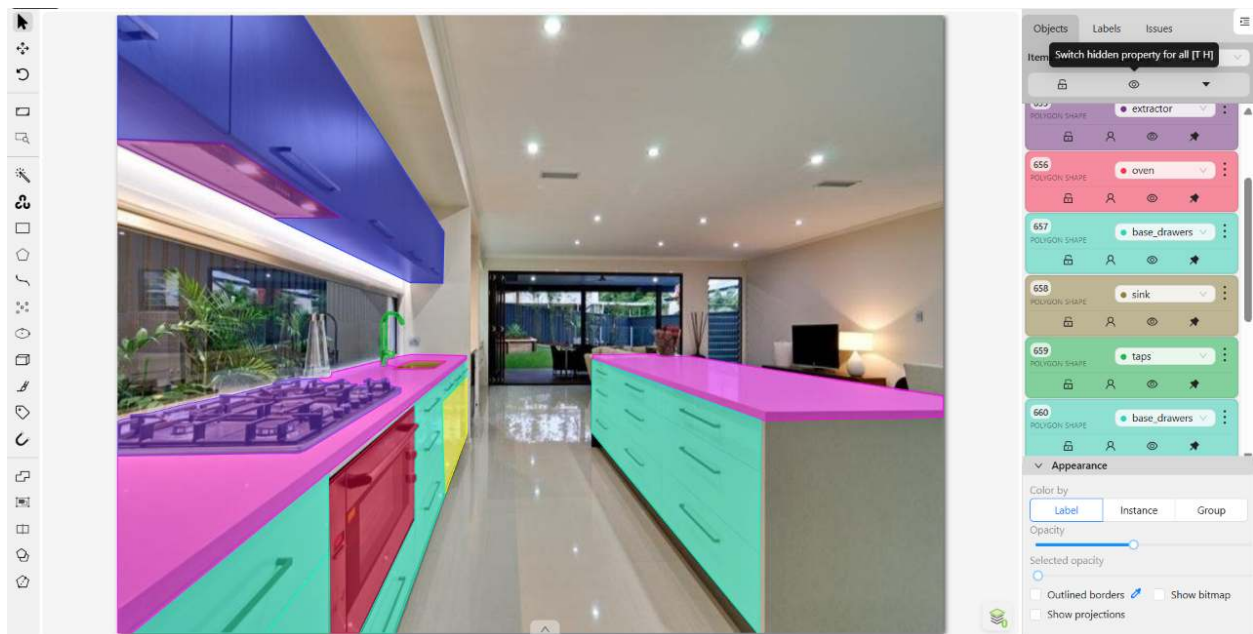
[Fig 5.21]: Before Annotations



[Fig 5.22]: After Annotations



[Fig 5.23]: Before Annotations



[Fig 5.24]: After Annotations



[Fig 5.25]: Before Annotations



[Fig 5.26]: After Annotations





[Fig 5.27]: Before Annotations



[Fig 5.28]: After Annotations





[Fig 5.29]: Before Annotations



[Fig 5.30]: After Annotations



[Fig 5.31]: Before Annotations



[Fig 5.32]: After Annotations

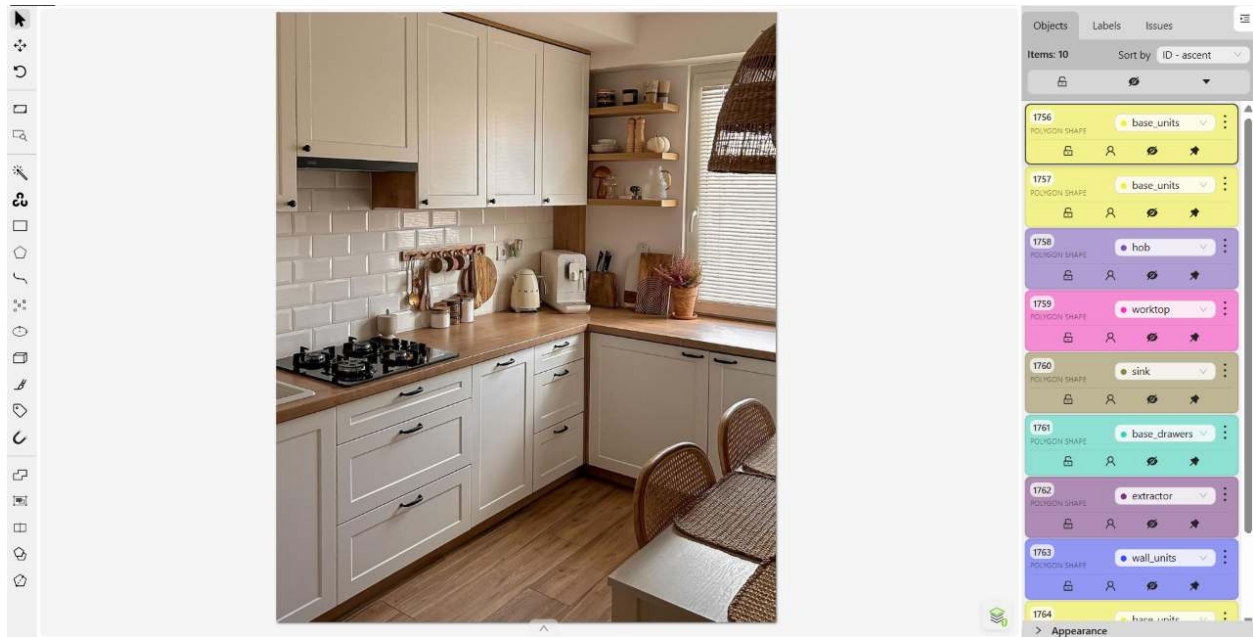


[Fig 5.33]: Before Annotations

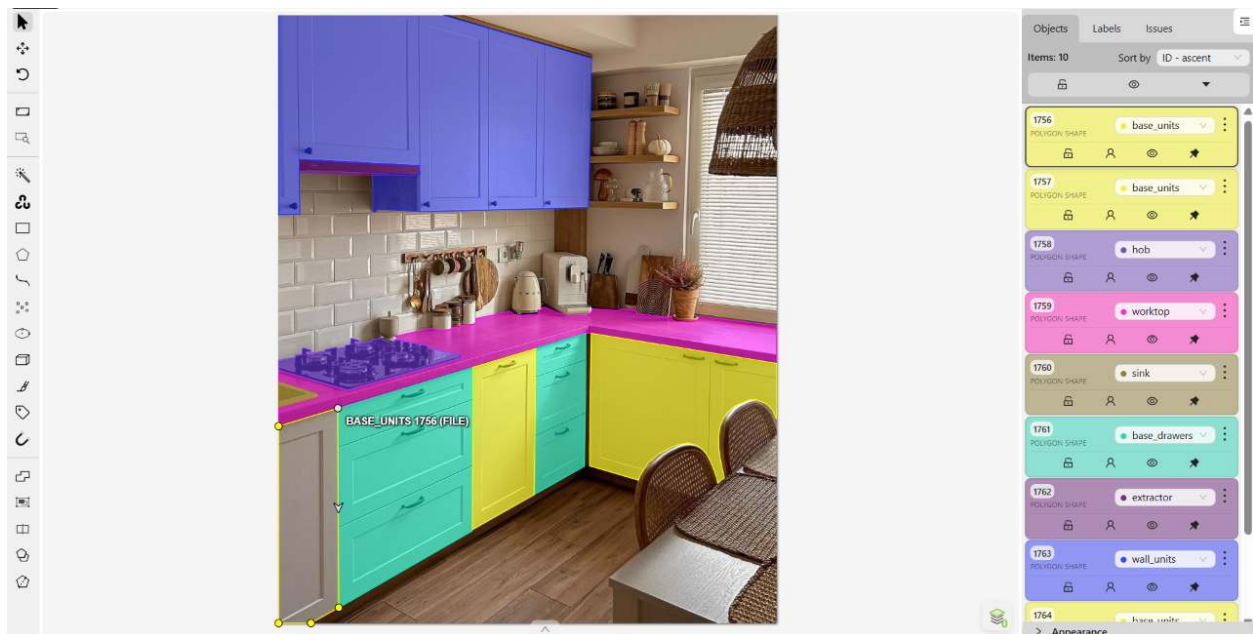


[Fig 5.34]: After Annotations





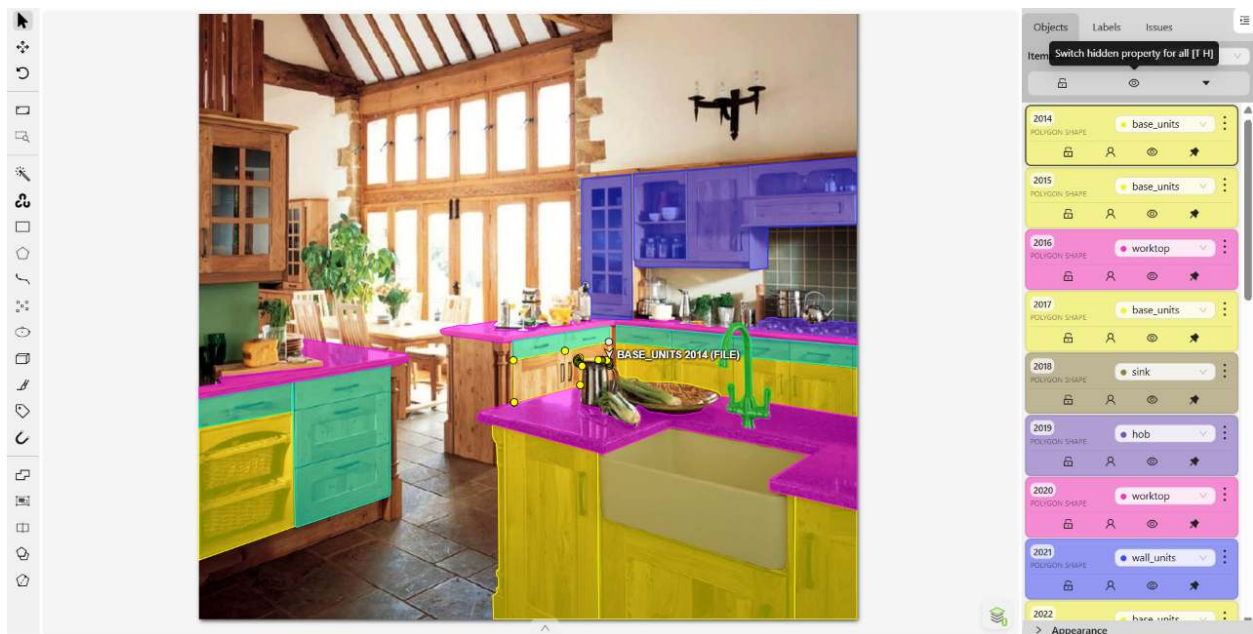
[Fig 5.35]: Before Annotations



[Fig 5.36]: After Annotations



[Fig 5.37]: Before Annotations



[Fig 5.38]: After Annotations





[Fig 5.39]: Before Annotations



[Fig 5.40]: After Annotations

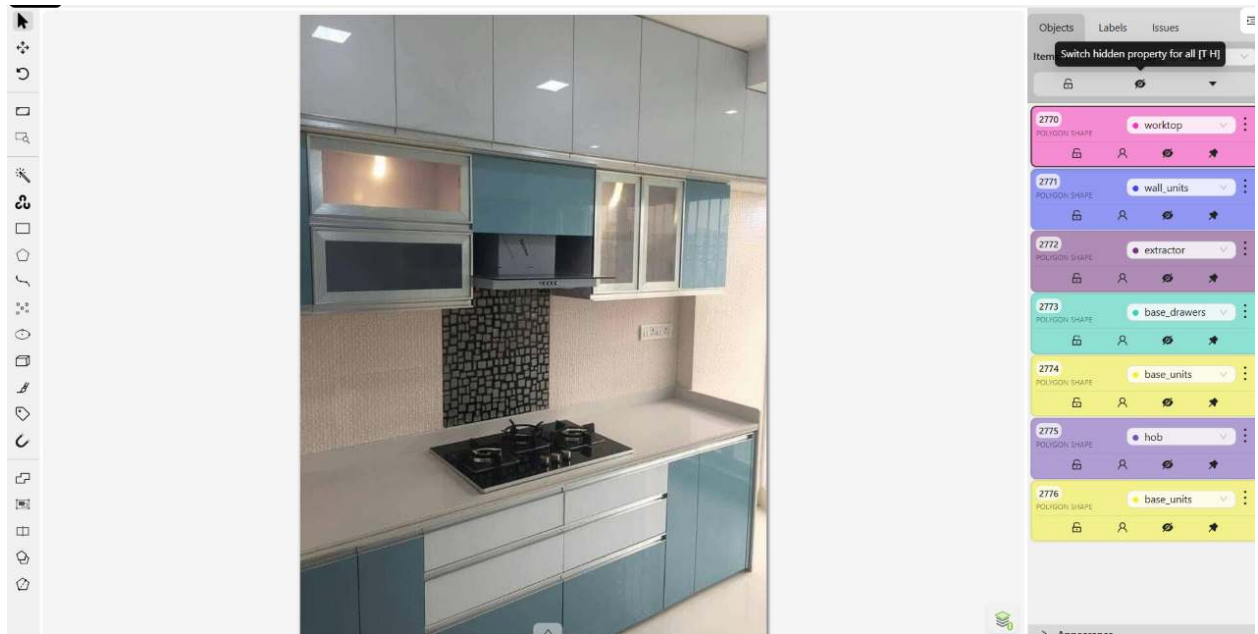




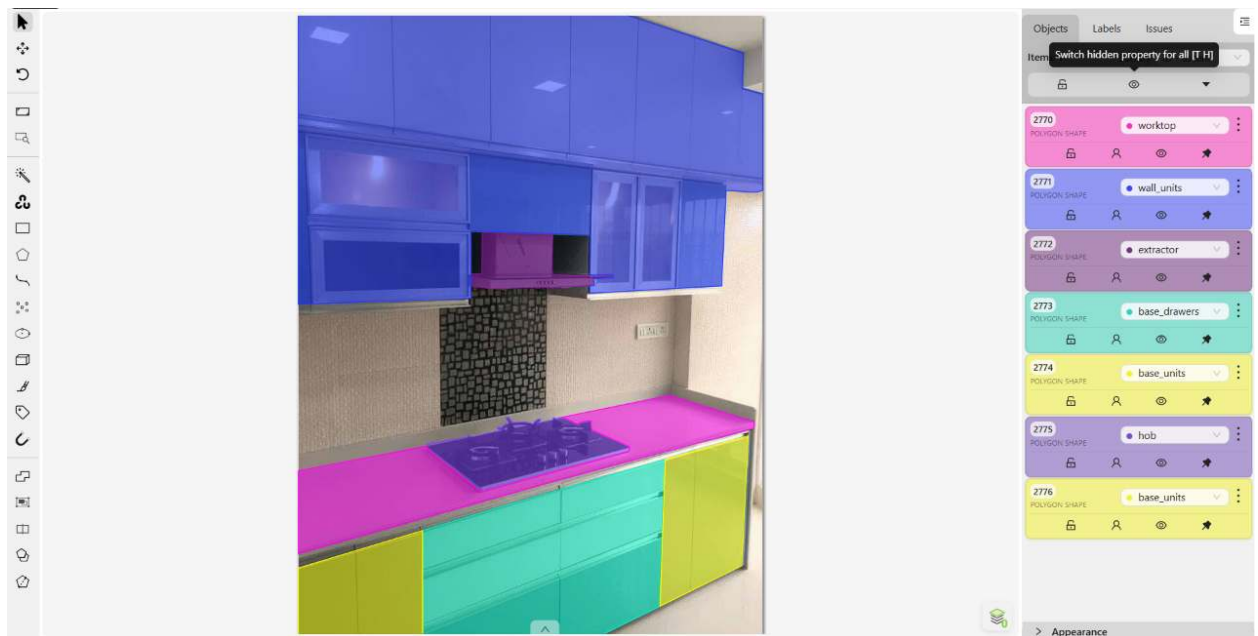
[Fig 5.41]: Before Annotations



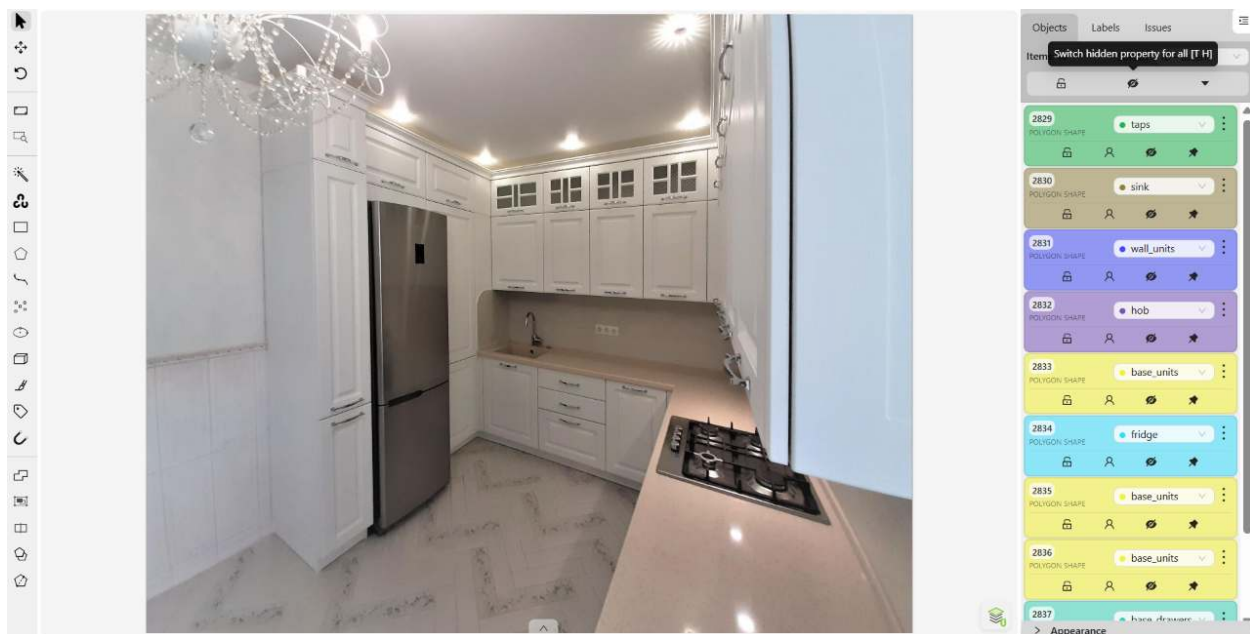
[Fig 5.42]: After Annotations



[Fig 5.43]: Before Annotations



[Fig 5.44]: After Annotations



[Fig 5.45]: Before Annotations



[Fig 5.46]: After Annotations







[Fig 5.49]: Before Annotations



[Fig 5.50]: After Annotations





[Fig 5.51]: Before Annotations



[Fig 5.52]: After Annotations

